

AN EXTENSION TO COMBINATORIAL COEFFICIENT CALCULUS

q-Binomial Coefficient Calculus

Gaussian weights, Bell expansions,
interpolation, reversion, and root-of-unity jets

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Abstract

Gaussian binomial coefficients fit into coefficient calculus in several different, noninterchangeable ways: as elementary and complete symmetric functions on geometric alphabets, as convolution structure constants, as interpolation matrices on q -integer nodes, and as polynomials whose local expansions are governed by Bernoulli, Stirling, and Bell numbers. This article develops those connections with explicit hypotheses and proofs. A finite partition-sum kernel evaluates arbitrary powers of products of q -shifted factorials, their compositions, and their Lagrange–Bürmann inverses. Further results include weighted q -binomial inversion, two Newton interpolation formulas, a normalization-specific q -Stirling calculus, ordinary composition in q -divided-power coordinates, coupled two-variable inversion, exact geometric extrapolation, and formulas for every derivative of a Gaussian polynomial at every root of unity. Classical identities are distinguished from the organizing synthesis. In particular, a change of factorial normalization is not confused with a Jackson chain rule, and distinct q -Catalan constructions are not identified. A reproducible exact-arithmetic verification program accompanies the source and rendered article; its finite tests supplement, rather than replace, the proofs.

Scope. A self-contained mathematical extension, not a modification of the repository and not a claim of new Lean formalization. All coefficient formulas have explicitly finite expansions at each prescribed degree.

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1 Purpose, source alignment, and algebraic conventions

1.1 What is being extended

The canonical *Combinatorial Coefficient Calculus* manuscript in the ProveIt repository [1] connects polynomial basis changes, Bell-polynomial composition, binomial inversion, and Lagrange–Bürmann/Good inversion. The present article extends these mechanisms to Gaussian coefficients. It uses the canonical T_EX source consulted on September 4, 2026, rather than assuming that the repository’s retained PDF has current source parity; the provenance document explicitly warns against that assumption [2].

The point is not to replace every $n!$ by $[n]_q!$. There are three separate operations. One may weight an ordinary combinatorial object by a statistic, change the basis used to record an ordinary formal series, or replace the derivative by a difference operator. Each produces useful q -formulas, but they need not produce the same formulas. Keeping these operations separate is essential for correct composition and inversion.

The classical q -binomial and terminating summation identities are standard [3, 4, 5]. They are proved here because their proofs reveal the coefficient mechanisms used later. The principal contribution of this extension is the explicit integration of these identities with the source’s coefficient calculus, together with worked finite-sum consequences. No historical priority claim is made for identities obtained by combining standard tools.

1.2 Coefficient rings and boundary values

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Unless stated otherwise, q is an indeterminate and computations take place in a commutative $\mathbb{Q}(q)$ -algebra R . Additional commuting parameters may be adjoined. The main series variable is z , so $R[[z]]$ means formal power series in z , not in q . An identity in $\mathbb{Z}[q]$ can subsequently be specialized at any complex q , including a root of unity. A rational expression can be specialized only where it is defined, or after its removable singularities have been eliminated.

Write

$$[n]_q = 1 + q + \dots + q^{n-1}, \quad [n]_q! = \prod_{j=1}^n [j]_q, \quad D_n = (q; q)_n = (1 - q)^n [n]_q!, \quad (1.1)$$

with $[0]_q = 0$, $[0]_q! = D_0 = 1$. The shifted factorial and Gaussian coefficient are

$$(a; q)_N = \prod_{j=0}^{N-1} (1 - aq^j), \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{D_n}{D_k D_{n-k}} \quad (0 \leq k \leq n). \quad (1.2)$$

For $n \geq 0$, put $\begin{bmatrix} n \\ k \end{bmatrix}_q = 0$ when $k < 0$ or $k > n$. Negative upper entries, when used, will receive a separate definition. Empty products are 1, empty sums are 0, and $0^0 = 1$ in finite coefficient formulas. Thus every zeroth coefficient below is unambiguous. A coefficient of negative degree is zero.

A series A with $A(0) = 1$ has the formal powers

$$A^\alpha = \exp(\alpha \log A), \quad \log(1 + u) = \sum_{j \geq 1} \frac{(-1)^{j-1}}{j} u^j. \quad (1.3)$$

These require a $\mathbb{Q}$ -algebra, but no analytic branch. Analytic interpretations are stated separately in [section 16](#).

1.3 The ordinary Bell kernel retained throughout

For $n \geq 0$ let

$$\mathcal{P}(n) = \{(m_1, \dots, m_n) \in \mathbb{N}^n : \sum_{j=1}^n j m_j = n\}.$$

For $n = 0$ this consists of the empty tuple. Define

$$\mathcal{E}_n(L) = \sum_{m \in \mathcal{P}(n)} \prod_{j=1}^n \frac{(L_j/j)^{m_j}}{m_j!}. \quad (1.4)$$

The complete exponential Bell polynomial B_n is normalized by

$$\exp\left(\sum_{j \geq 1} x_j \frac{z^j}{j!}\right) = \sum_{n \geq 0} B_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (1.5)$$

Consequently

$$\mathcal{E}_n(L) = \frac{1}{n!} B_n(0!L_1, 1!L_2, \dots, (n-1)!L_n), \quad \exp\left(\sum_{j \geq 1} \frac{L_j}{j} z^j\right) = \sum_{n \geq 0} \mathcal{E}_n(L) z^n. \quad (1.6)$$

Both statements follow by multiplying the individual exponential series. In particular, (1.4) is a definition by a finite sum, not by a recurrence. The first values provide a useful normalization check:

$$\begin{aligned} \mathcal{E}_0 &= 1, & \mathcal{E}_1 &= L_1, & \mathcal{E}_2 &= \frac{L_1^2 + L_2}{2}, \\ \mathcal{E}_3 &= \frac{L_1^3 + 3L_1L_2 + 2L_3}{6}, & \mathcal{E}_4 &= \frac{L_1^4 + 6L_1^2L_2 + 3L_2^2 + 8L_1L_3 + 6L_4}{24}. \end{aligned} \quad (1.7)$$

The q -dependence will usually enter through the arguments L_j , not through a new Bell-polynomial definition.

2 Gaussian coefficients as coefficient extractors

2.1 The finite theorem, polynomiality, and symmetry

Theorem 2.1 (Finite q -binomial theorem). *For $N \geq 0$,*

$$\prod_{j=0}^{N-1} (1 + zq^j) = \sum_{k=0}^N q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k. \quad (2.1)$$

Moreover,

$$\begin{bmatrix} N \\ k \end{bmatrix}_q = \sum_{0 \leq i_1 < \dots < i_k < N} q^{i_1 + \dots + i_k - \binom{k}{2}} \in \mathbb{Z}_{\geq 0}[q]. \quad (2.2)$$

The $k = 0$ sum has one empty term.

Proof. Expanding the product gives the subset sum before division by $q^{\binom{k}{2}}$. Let $A_{N,k}$ denote that normalized sum. Separating subsets that contain $N-1$ from those that do not gives

$$A_{N,k} = A_{N-1,k} + q^{N-k} A_{N-1,k-1}.$$

The quotient in (1.2) satisfies the same identity: after using the common denominator $D_k D_{N-k}$, the numerator is $D_{N-1}((1 - q^{N-k}) + q^{N-k}(1 - q^k)) = D_N$. The boundary values agree, so induction proves the identification. Finally, $i_j \geq j-1$, hence every exponent in (2.2) is nonnegative. $\square$

Corollary 2.2 (Pascal, symmetry, reciprocity, and classical limit). *For the indicated nonnegative indices,*

$$\begin{bmatrix} N \\ k \end{bmatrix}_q = \begin{bmatrix} N-1 \\ k \end{bmatrix}_q + q^{N-k} \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q = q^k \begin{bmatrix} N-1 \\ k \end{bmatrix}_q + \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q, \quad (2.3)$$

$$\begin{bmatrix} N \\ k \end{bmatrix}_q = \begin{bmatrix} N \\ N-k \end{bmatrix}_q, \quad \begin{bmatrix} N \\ k \end{bmatrix}_{q^{-1}} = q^{-k(N-k)} \begin{bmatrix} N \\ k \end{bmatrix}_q, \quad (2.4)$$

$$\begin{bmatrix} N \\ k \end{bmatrix}_1 = \binom{N}{k}, \quad \begin{bmatrix} N \\ k \end{bmatrix}_0 = 1 \quad (0 \leq k \leq N). \quad (2.5)$$

For $0 \leq k \leq N$ the degree is $k(N-k)$ and the leading and constant coefficients are 1.

Proof. The first Pascal identity was proved above. Symmetry follows from the factorial quotient and gives the second Pascal identity. Replacing each $1 - q^{-j}$ by $-q^{-j}(1 - q^j)$ gives reciprocity. At $q = 1$, the subset sum counts k -subsets. Its smallest exponent, zero, occurs once, and its largest exponent, $k(N-k)$, occurs once. This also proves the assertion at zero. $\square$

Put $\lambda_j = i_j - (j-1)$. Then $0 \leq \lambda_1 \leq \dots \leq \lambda_k \leq N-k$, and the exponent in (2.2) is $\sum_j \lambda_j$. Thus $\begin{bmatrix} N \\ k \end{bmatrix}_q$ enumerates partitions inside a $k \times (N-k)$ rectangle by size; see also [6]. This interpretation will justify positivity of coefficients that later emerge from alternating sums.

2.2 Complete symmetric functions and negative upper entries

Theorem 2.3 (Reciprocal finite product). *For $N \geq 1$,*

$$\frac{1}{(z; q)_N} = \sum_{r \geq 0} \begin{bmatrix} N+r-1 \\ r \end{bmatrix}_q z^r. \quad (2.6)$$

Equivalently,

$$\begin{bmatrix} N+r-1 \\ r \end{bmatrix}_q = \sum_{\substack{a_0 + \dots + a_{N-1} = r \\ a_j \geq 0}} q^{\sum_{j=0}^{N-1} j a_j} = \sum_{0 \leq i_1 \leq \dots \leq i_r < N} q^{i_1 + \dots + i_r}. \quad (2.7)$$

Proof. The first finite sum comes from expanding every geometric series. A weakly increasing r -tuple is carried to a strictly increasing tuple by $i_j \mapsto i_j + j - 1$. Its entries lie in $\{0, \dots, N+r-2\}$ and its weight increases by $\binom{r}{2}$. Applying (2.2) proves the result. $\square$

For $N = 0$, the reciprocal product is 1; this case is not encoded by a Gaussian coefficient with upper entry $r-1$.

For $M \geq 1$ and $r \geq 0$ define a negative upper entry by the finite product

$$\begin{bmatrix} -M \\ r \end{bmatrix}_q := \prod_{j=0}^{r-1} \frac{1 - q^{-M-j}}{1 - q^{j+1}}.$$

Factoring each numerator gives the explicit conversion

$$\begin{bmatrix} -M \\ r \end{bmatrix}_q = (-1)^r q^{-Mr - \binom{r}{2}} \begin{bmatrix} M+r-1 \\ r \end{bmatrix}_q. \quad (2.8)$$

Only the upper index has been extended here. This avoids the competing conventions that arise when both indices are negative; a broader treatment is available in [8].

2.3 Gaussian multinomials and a finite-field interpretation

For nonnegative $k_1 + \dots + k_s = N$ set

$$\left[\begin{matrix} N \\ k_1, \dots, k_s \end{matrix} \right]_q = \frac{D_N}{D_{k_1} \dots D_{k_s}}.$$

It is a polynomial because it factors as $\left[\begin{matrix} N \\ k_1 \end{matrix} \right]_q \left[\begin{matrix} N-k_1 \\ k_2 \end{matrix} \right]_q \dots \left[\begin{matrix} k_s \\ k_s \end{matrix} \right]_q$. When $q = Q$ is a prime power, $\left[\begin{matrix} N \\ k \end{matrix} \right]_Q$ counts k -dimensional subspaces of $\mathbb{F}_Q^N$: count ordered linearly independent k -tuples and divide by the number of ordered bases of one such subspace, obtaining

$$\frac{\prod_{j=0}^{k-1} (Q^N - Q^j)}{\prod_{j=0}^{k-1} (Q^k - Q^j)} = \left[\begin{matrix} N \\ k \end{matrix} \right]_Q.$$

Successively choosing subspaces of prescribed quotient dimensions gives the multinomial interpretation as a flag count. This is a distinct interpretation from weighting subsets; both evaluate the same polynomial.

3 Convolution and terminating summation identities

3.1 Two-block and multiblock Vandermonde identities

Theorem 3.1 (*q*-Vandermonde). For $M, N, r \geq 0$,

$$\left[\begin{matrix} M+N \\ r \end{matrix} \right]_q = \sum_k q^{(M-k)(r-k)} \left[\begin{matrix} M \\ k \end{matrix} \right]_q \left[\begin{matrix} N \\ r-k \end{matrix} \right]_q. \quad (3.1)$$

The sum is over $\max(0, r-N) \leq k \leq \min(M, r)$. More generally, if $N_1 + \dots + N_s = N$, then

$$\left[\begin{matrix} N \\ r \end{matrix} \right]_q = \sum_{\substack{k_1 + \dots + k_s = r \\ 0 \leq k_i \leq N_i}} q^{\sum_{i < j} (N_i - k_i)k_j} \prod_{i=1}^s \left[\begin{matrix} N_i \\ k_i \end{matrix} \right]_q. \quad (3.2)$$

Proof. Split $(-z; q)_{M+N}$ as $(-z; q)_M (-zq^M; q)_N$ and use (2.1). The coefficient with k choices from the first block is $q^{\binom{k}{2} + \binom{r-k}{2} + M(r-k)} \left[\begin{matrix} M \\ k \end{matrix} \right]_q \left[\begin{matrix} N \\ r-k \end{matrix} \right]_q$. Divide by $q^{\binom{r}{2}}$ to obtain (3.1). For several blocks, their initial exponents are $0, N_1, N_1 + N_2, \dots$. The same calculation subtracts $\sum_{i < j} k_i k_j$ from $\sum_{i < j} N_i k_j$, proving (3.2). $\square$

Corollary 3.2 (Three useful companions). For $M, N \geq 1$ and $r \geq 0$,

$$\left[\begin{matrix} M+N+r-1 \\ r \end{matrix} \right]_q = \sum_{k=0}^r q^{M(r-k)} \left[\begin{matrix} M+k-1 \\ k \end{matrix} \right]_q \left[\begin{matrix} N+r-k-1 \\ r-k \end{matrix} \right]_q, \quad (3.3)$$

$$\delta_{r0} = \sum_{k=0}^{\min(N,r)} (-1)^k q^{\binom{k}{2}} \left[\begin{matrix} N \\ k \end{matrix} \right]_q \left[\begin{matrix} N+r-k-1 \\ r-k \end{matrix} \right]_q. \quad (3.4)$$

For $0 \leq k \leq n$,

$$\sum_{j=k}^n q^{j-k} \left[\begin{matrix} j \\ k \end{matrix} \right]_q = \left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right]_q. \quad (3.5)$$

Proof. The first formula compares coefficients in $1/(z; q)_{M+N} = 1/((z; q)_M (zq^M; q)_N)$. The second compares coefficients in $(z; q)_N / (z; q)_N = 1$. For the third, rearrange the first Pascal identity as $q^{j-k} \left[\begin{matrix} j \\ k \end{matrix} \right]_q = \left[\begin{matrix} j+1 \\ k+1 \end{matrix} \right]_q - \left[\begin{matrix} j \\ k+1 \end{matrix} \right]_q$ and telescope. $\square$

3.2 The infinite theorem with a precise formal meaning

Theorem 3.3 (The *q*-binomial series). In $\mathbb{Q}(q, a)[[z]]$ define

$$R_a(z) = \exp \left(\sum_{j \geq 1} \frac{1 - a^j}{j(1 - q^j)} z^j \right). \quad (3.6)$$

Then

$$R_a(z) = \sum_{n \geq 0} \frac{(a; q)_n}{D_n} z^n = \frac{(az; q)_\infty}{(z; q)_\infty}. \quad (3.7)$$

The last expression denotes either its coefficientwise *q*-adic interpretation or, for $|q| < 1$ and $|z| < 1$, the convergent analytic product. It is not a limit of finite products in the discrete-coefficient *z*-adic topology alone.

Proof. Subtract the logarithm at *qz* from the logarithm at *z* in (3.6). Exponentiation gives

$$(1 - z)R_a(z) = (1 - az)R_a(qz).$$

If $c_n = [z^n]R_a$, this implies $(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}$ for $n \geq 1$, with $c_0 = 1$. Multiplying these finitely many ratios yields $c_n = (a; q)_n / D_n$. Expanding logarithms of the infinite products gives (3.6); this calculation is valid coefficientwise *q*-adically and in the indicated analytic domain. For finite products the coefficient of *z* contains $1 + q + \dots + q^{N-1}$, which does not eventually stabilize in $\mathbb{Q}(q)$. This proves the topology warning as well. $\square$

Proposition 3.4 (Cocycle and shifted-factorial convolution). One has $R_a(z)R_b(az) = R_{ab}(z)$ and therefore

$$\sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a; q)_k (b; q)_{n-k} a^{n-k} = (ab; q)_n. \quad (3.8)$$

Proof. In the logarithm the summands combine as $(1 - a^j) + a^j(1 - b^j) = 1 - (ab)^j$. Extracting z^n and multiplying by D_n proves the convolution. $\square$

Theorem 3.5 (The two terminating *q*-Chu–Vandermonde sums). For $n \geq 0$, as rational identities in generic parameters,

$$\sum_{k=0}^n \frac{(q^{-n}; q)_k (a; q)_k}{D_k(c; q)_k} q^k = a^n \frac{(c/a; q)_n}{(c; q)_n}, \quad (3.9)$$

$$\sum_{k=0}^n \frac{(q^{-n}; q)_k (a; q)_k}{D_k(c; q)_k} \left(\frac{cq^n}{a} \right)^k = \frac{(c/a; q)_n}{(c; q)_n}. \quad (3.10)$$

Specializations require the displayed denominators to be nonzero or a separately justified cancellation. In (3.9), the right side can be read as $\prod_{j=0}^{n-1} (a - cq^j) / (c; q)_n$, avoiding an artificial singularity at $a = 0$.

Proof. In (3.8) replace its parameters by A, b , interchange k and $n - k$, and divide by $(A; q)_n$. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{-n}; q)_k}{D_k} (-1)^k q^{nk - \binom{k}{2}}, \quad \frac{(A; q)_{n-k}}{(A; q)_n} = \frac{(-A)^{-k} q^{-nk + k(k+1)/2}}{(q^{1-n}/A; q)_k}.$$

The result is the left side of (3.9) with $b = a$ and $c = q^{1-n}/A$. Reversing the factors in the quotient on the right gives $(Aa; q)_n / (A; q)_n = a^n (c/a; q)_n / (c; q)_n$. For (3.10), replace (q, a, c) by (q^{-1}, a^{-1}, c^{-1}) in (3.9) and use

$$(x; q^{-1})_k = (-x)^k q^{-\binom{k}{2}} (x^{-1}; q)_k.$$

The powers remaining in the summand are $(cq^n/a)^k$; those on the right cancel. $\square$

4 The master product-to-Bell theorem

4.1 Geometric power sums

For a finite $N \geq 0$ define

$$S_N(t) = \sum_{j=0}^{N-1} t^j. \quad (4.1)$$

This polynomial, rather than $(1 - t^N)/(1 - t)$, is the preferred expression at $t = 1$. When $N = \infty$, write $S_\infty(t) = (1 - t)^{-1}$ with the formal or analytic interpretation of [theorem 3.3](#).

Theorem 4.1 (Ramified shifted-factorial coefficient calculus). *Let d_r, h_r be positive integers, $N_r \in \mathbb{N} \cup \{\infty\}$, and let a_r, α_r be commuting parameters, $1 \leq r \leq s$. Set*

$$P(z) = \prod_{r=1}^s (a_r z^{d_r}; q^{h_r})_{N_r}^{\alpha_r}, \quad (4.2)$$

and define, by finite sums for each $j \geq 1$,

$$L_j = - \sum_{\substack{1 \leq r \leq s \\ d_r | j}} d_r \alpha_r a_r^{j/d_r} S_{N_r}(q^{h_r j/d_r}). \quad (4.3)$$

Then for every $n \geq 0$ and every scalar β ,

$$[z^n] P(z)^\beta = \mathcal{E}_n(\beta L) = \sum_{m \in \mathcal{P}(n)} \prod_{j=1}^n \frac{(\beta L_j/j)^{m_j}}{m_j!}. \quad (4.4)$$

Here infinite factors are defined by their logarithms over $\mathbb{Q}(q)$; finite factors need no such qualification.

Proof. For one finite factor,

$$\log(a z^d; q^h)_N = - \sum_{\ell \geq 1} \frac{a^\ell}{\ell} S_N(q^{h\ell}) z^{d\ell}.$$

The same formula defines an infinite factor with S_∞ . In degree j , a factor contributes only if $d \mid j$; writing $\ell = j/d$ makes its contribution $-da^{j/d} S_N(q^{hj/d})/j$. Summing the exponents α_r yields $\log P = \sum_{j \geq 1} L_j z^j/j$. Equation (1.6) proves (4.4). All coefficient extractions use only $L_1, \dots, L_n$. $\square$

Corollary 4.2 (Gaussian coefficients as explicit Bell sums). *For $N \geq 1$ and $k \geq 0$,*

$$q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q = \sum_{m \in \mathcal{P}(k)} \prod_{j=1}^k \frac{1}{m_j!} \left(\frac{(-1)^{j-1} S_N(q^j)}{j} \right)^{m_j}, \quad (4.5)$$

$$\begin{bmatrix} N+k-1 \\ k \end{bmatrix}_q = \sum_{m \in \mathcal{P}(k)} \prod_{j=1}^k \frac{1}{m_j!} \left(\frac{S_N(q^j)}{j} \right)^{m_j}. \quad (4.6)$$

The first expression is zero for $k > N$.

Proof. Apply [theorem 4.1](#) to $(-z; q)_N$ and $1/(z; q)_N$, respectively, and compare with [theorems 2.1](#) and [2.3](#). $\square$

These are the geometric-alphabet specializations of the source's symmetric-function/Bell correspondence. No q -Bell polynomial is needed. At $q = 1$ they become finite Bell-sum formulas for $\binom{N}{k}$ and $\binom{N+k-1}{k}$.

4.2 Composition, arbitrary powers, and parameter derivatives

Corollary 4.3 (Composition with a unit product). *If $H(u) = \sum_{k \geq 0} h_k u^k$ and P is as in theorem 4.1, then*

$$[z^n]H(P(z) - 1) = \sum_{k=0}^n h_k \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} \mathcal{E}_n(jL). \quad (4.7)$$

If $M(\mathbf{m}) = \sum_j m_j$, then

$$[z^n]P(z)^\beta (\log P(z))^r = \sum_{\mathbf{m} \in \mathcal{P}(n)} (M(\mathbf{m}))_r \beta^{M(\mathbf{m})-r} \prod_{j=1}^n \frac{(L_j/j)^{m_j}}{m_j!}, \quad (4.8)$$

where terms with $M(\mathbf{m}) < r$ are zero, so no negative power of β is used.

Proof. Expand $(P - 1)^k = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} P^j$. Since $P - 1$ has positive z -order, only $k \leq n$ can contribute. For the second formula differentiate (4.4) r times in β ; its degree in β is at most n . $\square$

For example, put

$$P(z) = \frac{(-z^2; q^2)_M^\gamma}{(az; q)_N^\alpha}.$$

The logarithmic inputs are explicitly

$$L_j = \alpha a^j S_N(q^j) - \mathbf{1}_{2|j} 2\gamma (-1)^{j/2} S_M(q^j). \quad (4.9)$$

Thus all powers, logarithmic factors, and compositions of this mixed even/ordinary product are evaluated by one finite kernel. This example also makes clear why the ramification factor d_r in (4.3) cannot be omitted.

5 Extracting coefficients in the base parameter

The preceding sections treated q as a parameter. We now extract powers of q itself. These are different coefficient questions.

Theorem 5.1 (Divisor-sum formula for Gaussian coefficients). *Fix $0 \leq k \leq n$ and put*

$$A_m(n, k) = \sum_{\substack{1 \leq d \leq k \\ d|m}} d - \sum_{\substack{n-k+1 \leq d \leq n \\ d|m}} d. \quad (5.1)$$

Then, for every $r \geq 0$,

$$[q^r] \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{\mathbf{m} \in \mathcal{P}(r)} \prod_{j=1}^r \frac{(A_j(n, k)/j)^{m_j}}{m_j!}. \quad (5.2)$$

In particular the expression vanishes for $r > k(n - k)$.

Proof. As a unit of $\mathbb{Q}[[q]]$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{\prod_{d=n-k+1}^n (1 - q^d)}{\prod_{d=1}^k (1 - q^d)}.$$

The coefficient of q^m in its logarithm is $A_m(n, k)/m$, since $[q^m] \log(1 - q^d) = -d/m$ when $d \mid m$ and is zero otherwise. Apply (1.6), now with q in place of z . The degree bound was proved in theorem 2.2. $\square$

Proposition 5.2 (Bounded-part inclusion–exclusion). *Let*

$$p_{\leq k}(r) = \#\{(a_1, \dots, a_k) \in \mathbb{N}^k : \sum_{j=1}^k ja_j = r\},$$

with value zero for $r < 0$, and $p_{\leq 0}(r) = \delta_{r0}$. Then

$$[q^r] \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{S \subseteq \{1, \dots, k\}} (-1)^{|S|} p_{\leq k} \left(r - \sum_{i \in S} (n - k + i) \right). \quad (5.3)$$

Proof. Expand the numerator of the product in the preceding proof over subsets S . The reciprocal denominator has coefficient $p_{\leq k}(r)$ by the geometric series expansion. Their Cauchy product gives the formula. $\square$

For example,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = 1 + q + 2q^2 + q^3 + q^4. \quad (5.4)$$

Both (5.2) and (5.3) recover these coefficients without a Gaussian recurrence. Their cancellations above degree four are exact identities. The subset formula, not the alternating presentation, explains why the surviving coefficients are nonnegative integers.

6 Gaussian convolution and the correct inverse transform

6.1 Divided-power coordinates and two exponentials

Define

$$e_q(z) = \sum_{n \geq 0} \frac{z^n}{[n]_q!}, \quad E_q(z) = \sum_{n \geq 0} q^{\binom{n}{2}} \frac{z^n}{[n]_q!}. \quad (6.1)$$

Using theorem 3.3 with $a = 0$ and the finite theorem followed by its coefficientwise limit gives

$$e_q(z) = \frac{1}{((1-q)z; q)_\infty}, \quad E_q(z) = (-(1-q)z; q)_\infty, \quad e_q(z)E_q(-z) = 1. \quad (6.2)$$

For completeness, the coefficientwise limit used for the second equality is $q^{\binom{n}{2}}/D_n$: in the finite theorem, $[N]_q = \prod_{j=0}^{n-1} (1 - q^{N-j})/D_n$ tends q -adically to $1/D_n$. Thus (6.2) also holds purely formally over $\mathbb{Q}(q)$. The functions in (6.1) use q -factorials; a convention with D_n in the denominator rescales their argument by $1 - q$.

For sequences a, b put

$$(a \star_q b)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k b_{n-k}, \quad \mathcal{G}_q(a; z) = \sum_{n \geq 0} a_n \frac{z^n}{[n]_q!}. \quad (6.3)$$

Multiplication of formal series immediately proves

$$\mathcal{G}_q(a \star_q b; z) = \mathcal{G}_q(a; z) \mathcal{G}_q(b; z). \quad (6.4)$$

Hence $\star_q$ is associative and commutative and has identity $\epsilon_n = \delta_{n0}$. Alternatively, associativity follows directly from $\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q$, so it is a polynomial identity and does not depend on the invertibility of q -factorials after specialization.

Theorem 6.1 (Weighted Gaussian inversion). *Let c be a scalar. The two transforms*

$$b_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q c^{n-k} a_k, \quad (6.5)$$

$$a_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q c^{n-k} b_k \quad (6.6)$$

are mutually inverse. These are polynomial identities, valid over every commutative ring after assigning a value to q and c .

Proof. Over $\mathbb{Q}(q)$, the forward transform multiplies the generating series by $e_q(cz)$ and the inverse multiplies by $E_q(-cz)$. Their product is 1. For a denominator-free proof, the coefficient of a_j after both transforms is

$$c^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_q \sum_{r=0}^{n-j} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n-j \\ r \end{bmatrix}_q = c^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_q (1; q)_{n-j}.$$

This is 1 when $n = j$ and 0 otherwise. Interchanging the transforms gives the same sum, proving both inverse compositions. Since the entire argument is a polynomial identity with integer coefficients, arbitrary specialization follows. $\square$

6.2 Why ordinary translation fails

Let T_c denote (6.5). Define the homogeneous polynomial

$$H_r(c, d; q) = \sum_{j=0}^r \begin{bmatrix} r \\ j \end{bmatrix}_q c^j d^{r-j}.$$

Associativity of convolution gives the exact composition law

$$(T_c T_d a)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q H_{n-k}(c, d; q) a_k. \quad (6.7)$$

For example,

$$H_2(c, d; q) = c^2 + (1+q)cd + d^2,$$

which is not $(c+d)^2$ for generic commuting c, d . Thus $T_c T_d \neq T_{c+d}$ in general, and T_{-c} is not the inverse of T_c . The missing factor in the naive inverse is precisely $q^{\binom{n-k}{2}}$. This contrasts with the ordinary weighted-translation theorem in the base article. The correct replacement is either (6.7), or the multiplicative cocycle of [theorem 3.4](#) when shifted-factorial kernels are used.

6.3 Subspace-lattice Möbius inversion

For a prime power Q , let $U \subseteq V$ be finite-dimensional vector spaces and $r = \dim(V/U)$. The Möbius function of their subspace interval is

$$\mu(U, V) = (-1)^r Q^{\binom{r}{2}}. \quad (6.8)$$

Indeed, the sum of this expression over intermediate subspaces of an r -dimensional quotient is $\sum_{j=0}^r \begin{bmatrix} r \\ j \end{bmatrix}_Q (-1)^j Q^{\binom{j}{2}} = (1; Q)_r$. It is 1 for $r = 0$ and zero otherwise, exactly the inverse-to-zeta condition. Grouping subspace sums by dimension produces [theorem 6.1](#) with $c = 1$. Thus the extra Gaussian inverse weight has a structural explanation: the Boolean lattice has Möbius weight $(-1)^r$, whereas the subspace lattice has the additional factor $Q^{\binom{r}{2}}$.

7 Newton interpolation and a q -Stirling basis change

7.1 The q -integer lattice: the deformation of ordinary Newton expansion

Define the monic polynomials

$$F_k(x) = \prod_{i=0}^{k-1} (x - [i]_q), \quad F_0(x) = 1. \quad (7.1)$$

Their normalized versions

$$N_k(x; q) = \frac{F_k(x)}{q^{\binom{k}{2}} [k]_q!}$$

satisfy $N_k([m]_q; q) = \begin{bmatrix} m \\ k \end{bmatrix}_q$ for $m \geq 0$. To see this, use $[m]_q - [i]_q = q^i [m-i]_q$ and multiply over i . This basis belongs to the broader setting of quantum integer-valued polynomials [9]; its normalization is stated here explicitly.

Theorem 7.1 (Finite q -Newton expansion). *For every polynomial p of degree at most d over R ,*

$$p(x) = \sum_{k=0}^d c_k N_k(x; q), \quad (7.2)$$

$$c_k = \sum_{j=0}^k (-1)^{k-j} q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q p([j]_q). \quad (7.3)$$

At $q = 1$ this becomes $p(x) = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}$.

Proof. The distinct degrees and unit leading coefficients of $N_0, \dots, N_d$ make them a basis. Evaluate its expansion at $x = [m]_q$ for $0 \leq m \leq d$: $p([m]_q) = \sum_{k=0}^m \begin{bmatrix} m \\ k \end{bmatrix}_q c_k$. Gaussian inversion gives (7.3). At $q = 1$, the factors in N_k become $(x - i)$, their denominator becomes $k!$, and the sum for c_k is the ordinary finite-difference formula. All denominators in this specialization are nonzero. $\square$

7.2 A specified pair of q -Stirling arrays

Definition 7.2. Define $S_q(n, k)$ and $s_q(n, k)$ by

$$x^n = \sum_{k=0}^n S_q(n, k) F_k(x), \quad F_n(x) = \sum_{k=0}^n s_q(n, k) x^k. \quad (7.4)$$

These definitions, rather than the words “ q -Stirling numbers” alone, fix the convention. Both arrays are zero outside $0 \leq k \leq n$ and have entry 1 at $(0, 0)$.

Theorem 7.3 (Finite formulas and inverse matrices). *For $n, k \geq 0$ with $k \leq n$,*

$$S_q(n, k) = \frac{1}{q^{\binom{k}{2}} [k]_q!} \sum_{j=0}^k (-1)^{k-j} q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q ([j]_q)^n, \quad (7.5)$$

$$s_q(n, k) = (-1)^{n-k} \sum_{\substack{I \subseteq \{0, \dots, n-1\} \\ |I|=n-k}} \prod_{i \in I} [i]_q. \quad (7.6)$$

Furthermore,

$$\sum_{j=k}^n S_q(n, j) s_q(j, k) = \delta_{nk}, \quad \sum_{j=k}^n s_q(n, j) S_q(j, k) = \delta_{nk}. \quad (7.7)$$

The arrays belong to $\mathbb{Z}[q]$, although (7.5) is displayed as a rational function. In fact $S_q(n, k) \in \mathbb{Z}_{\geq 0}[q]$.

Proof. Apply theorem 7.1 to $p(x) = x^n$ to obtain the first formula. Expand the product F_n over subsets to obtain the second. Substitution of the two basis expansions into each other gives both matrix identities. To prove polynomiality without division, note that $xF_k = F_{k+1} + [k]_q F_k$. Starting from $x^0 = F_0$ and multiplying by x shows that every $S_q(n, k)$ is a sum of products of q -integers with nonnegative integer coefficients. This argument also yields the check

$$S_q(n+1, k) = S_q(n, k-1) + [k]_q S_q(n, k),$$

but the explicit formula (7.5) does not depend on using this recurrence to evaluate earlier rows. $\square$

A second finite expression is

$$S_q(n, k) = \sum_{\substack{a_1 + \dots + a_k = n-k \\ a_j \geq 0}} \prod_{j=1}^k ([j]_q)^{a_j}, \quad n \geq k. \quad (7.8)$$

To prove it, the coefficient sequence on the right has generating series $\prod_{j=1}^k (1 - [j]_q t)^{-1}$. The recurrence in the preceding proof implies, for $G_k(t) = \sum_{n \geq k} S_q(n, k) t^{n-k}$, that $(1 - [k]_q t) G_k(t) = G_{k-1}(t)$, with $G_0 = 1$; hence the products agree. When $k = 0$, (7.8) means δ_{n0} . All three formulas degenerate to the corresponding ordinary Stirling formulas at $q = 1$. For example, $S_q(3, 2) = 2 + q$.

7.3 Geometric nodes are different

Let $P_k(x) = \prod_{i=0}^{k-1} (x - q^i)$. For $q \neq 0$ with the relevant powers $q^0, \dots, q^d$ distinct, every polynomial p of degree at most d has

$$p(x) = \sum_{k=0}^d A_k P_k(x), \quad A_k = \sum_{j=0}^k \frac{(-1)^j q^{-kj+j(j+1)/2}}{D_j D_{k-j}} p(q^j). \quad (7.9)$$

Indeed, $P_k(q^m) = (-1)^k q^{\binom{k}{2}} D_k \begin{bmatrix} m \\ k \end{bmatrix}_q$; apply the same triangular inversion. Equivalently, A_k is the usual divided difference on the geometric nodes, because

$$\prod_{\substack{0 \leq i \leq k \\ i \neq j}} (q^j - q^i) = (-1)^j q^{jk-j(j+1)/2} D_j D_{k-j}.$$

This supplies an independent interpolation proof of (7.9).

As $q \rightarrow 1$, these nodes all coalesce at 1 and $A_k \rightarrow p^{(k)}(1)/k!$; thus the limiting expansion is Taylor's formula at 1, not ordinary Newton expansion on the integers. One algebraic proof of the limit is that the monic bases P_k tend coefficientwise to $(x-1)^k$; their finite triangular change-of-basis matrix has diagonal 1 and hence a continuously varying inverse. The distinction between the two lattices is important in applications.

8 Ordinary composition in q -divided-power coordinates

8.1 A normalized Bell family, with ordinary multiplicities

The ordinary partial Bell polynomial is

$$\hat{B}_{n,k}(a_1, a_2, \dots) = \sum_{\substack{\sum_j m_j = k \\ \sum_j j m_j = n}} \frac{k!}{\prod_j m_j!} \prod_j a_j^{m_j}.$$

It satisfies $[z^n](\sum_{j \geq 1} a_j z^j)^k = \hat{B}_{n,k}(a)$. Define $\hat{B}_{0,0} = 1$ and $\hat{B}_{n,0} = 0$ for $n > 0$.

Definition 8.1. For ordinary substitution of commuting series, define

$$\mathcal{B}_{n,k}^{(q)}(g) = \frac{[n]_q!}{[k]_q!} \sum_{\substack{\sum_j m_j = k \\ \sum_j j m_j = n}} \frac{k!}{\prod_j m_j!} \prod_j \left(\frac{g_j}{[j]_q!} \right)^{m_j}. \quad (8.1)$$

Boundary values are $\mathcal{B}_{0,0}^{(q)} = 1$ and $\mathcal{B}_{n,0}^{(q)} = 0$ for $n > 0$.

This is a coordinate-normalized Bell family; it is not asserted to coincide with any other family called “ q -Bell polynomials.” Its exact relation to the base article’s ordinary Bell polynomial is

$$\mathcal{B}_{n,k}^{(q)}(g) = \frac{[n]_q!}{[k]_q!} \hat{B}_{n,k} \left(\frac{g_1}{[1]_q!}, \frac{g_2}{[2]_q!}, \dots \right). \quad (8.2)$$

Theorem 8.2 (Composition rule). *If $F(z) = \sum_{k \geq 0} f_k z^k / [k]_q!$ and $G(z) = \sum_{j \geq 1} g_j z^j / [j]_q!$, then*

$$[n]_q! [z^n] F(G(z)) = \sum_{k=0}^n f_k \mathcal{B}_{n,k}^{(q)}(g). \quad (8.3)$$

At $q = 1$, $\mathcal{B}_{n,k}^{(q)}$ becomes the exponential partial Bell polynomial $B_{n,k}$.

Proof. Expand G^k using the ordinary multinomial theorem. Its multiplicity for the profile (m_j) is $k! / \prod_j m_j!$; there is no Gaussian weight because the factors commute and ordinary substitution is being used. Multiply by $f_k / [k]_q!$, sum over k , and multiply by $[n]_q!$. At $q = 1$, cancellation of $k!$ yields the usual exponential Bell profile $n! / \prod_j m_j! (j!)^{m_j}$. $\square$

The first nontrivial examples expose a frequent normalization error:

$$\mathcal{B}_{3,2}^{(q)} = \frac{2[3]_q}{[2]_q} g_1 g_2, \quad (8.4)$$

$$\mathcal{B}_{4,2}^{(q)} = \frac{2[4]_q}{[2]_q} g_1 g_3 + \frac{[4]_q [3]_q}{([2]_q)^2} g_2^2, \quad (8.5)$$

$$\mathcal{B}_{4,3}^{(q)} = \frac{3[4]_q}{[2]_q} g_1^2 g_2. \quad (8.6)$$

Also $\mathcal{B}_{n,1}^{(q)} = g_n$ and $\mathcal{B}_{n,n}^{(q)} = g_1^n$. The rational coefficient $2[3]_q / [2]_q$ is not $[3]_q$. It need not be a polynomial in q , and it may have a pole at a root of unity. This is expected: the coordinate system itself uses inverses of q -factorials.

8.2 Riordan matrices as a coordinate change

Theorem 8.3 (Normalized Riordan composition). *Let $A(0)$ and $G'(0)$ be units and $G(0) = 0$. Define*

$$\mathcal{R}_q(A, G)_{n,k} = \frac{[n]_q!}{[k]_q!} [z^n] A(z) G(z)^k. \quad (8.7)$$

Then

$$\mathcal{R}_q(A, G) \mathcal{R}_q(B, H) = \mathcal{R}_q(A \cdot B \circ G, H \circ G), \quad (8.8)$$

$$\mathcal{R}_q(A, G)^{-1} = \mathcal{R}_q\left(\frac{1}{A \circ \overline{G}}, \overline{G}\right), \quad (8.9)$$

where $\overline{G}$ is the ordinary compositional inverse.

Proof. The matrix acts on the coefficient vector of F by sending its series to $A(z)F(G(z))$. Applying this operation twice proves (8.8); substituting $H = \overline{G}$ and $B = 1/(A \circ \overline{G})$ proves (8.9). Equivalently, if $R(A, G)_{n,k} = [z^n] AG^k$, then $\mathcal{R}_q = DRD^{-1}$ for $D = \text{diag}([0]_q!, [1]_q!, \dots)$. Thus this family is a diagonal renormalization of ordinary Riordan matrices, not an unexplained new substitution law. $\square$

In particular the Gaussian transform is $\mathcal{R}_q(e_q(cz), z)$ and its inverse is $\mathcal{R}_q(E_q(-cz), z)$, recovering [theorem 6.1](#).

9 Jackson calculus and the noncommuting binomial theorem

Define the Jackson derivative and dilation by

$$D_q f(z) = \frac{f(z) - f(qz)}{(1-q)z}, \quad T_q f(z) = f(qz). \quad (9.1)$$

The numerator has zero constant term, so the quotient is a formal series. One has $D_q z^n = [n]_q z^{n-1}$, $D_q T_q = q T_q D_q$, and

$$D_q(fg) = (D_q f)g + (T_q f)(D_q g). \quad (9.2)$$

These follow by direct subtraction.

Theorem 9.1 (Iterated Jackson product rule). *For $n \geq 0$,*

$$D_q^n(fg)(z) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k}z) (D_q^{n-k} g)(z). \quad (9.3)$$

Proof. The case $n = 0$ is immediate. Differentiate each summand using (9.2). The derivative falling on its first factor contributes $q^{n-k}(D_q^{k+1}f)(q^{n-k}z)D_q^{n-k}g$; the derivative falling on its second factor contributes $(D_q^k f)(q^{n-k+1}z)D_q^{n-k+1}g$. After reindexing, the coefficient of the new k th term is $\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$ by Pascal's identity. $\square$

At the origin, $D_q^n f(0) = [n]_q! [z^n] f$. Consequently [theorem 8.2](#) gives an exact origin formula for $D_q^n(F \circ G)(0)$ when $G(0) = 0$. It does *not* imply an ordinary-looking Jackson chain rule away from zero. The valid first-order identity is

$$D_q(F \circ G)(z) = D_q G(z) F[G(z), G(qz)], \quad (9.4)$$

where $F[u, v] = (F(u) - F(v))/(u - v)$ is the divided difference. Its polynomial or formal expansion removes the apparent denominator when $u = v$. For $F(w) = w^2$ this is $D_q G(z)(G(z) + G(qz))$, not generally $[2]_q G(z) D_q G(z)$. For instance $G(z) = z^2$ makes the former second factor $(1 + q^2)z^2$.

There is, however, a genuine Gaussian binomial theorem for noncommuting variables.

Proposition 9.2 (Quantum-plane binomial theorem). *In an associative algebra with a central parameter q and $YX = qXY$,*

$$(X + Y)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q X^{n-k} Y^k. \quad (9.5)$$

Proof. Right-multiply by $X + Y$ and use $Y^k X = q^k X Y^k$. The new coefficient is $q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$. The initial case is $n = 0$. $\square$

This theorem is consistent with the failure of ordinary scalar translation in [section 6](#): the Gaussian cross term is produced by a commutation relation, not by the ordinary expansion of commuting $c + d$.

10 Lagrange inversion with shifted-factorial weights

10.1 A fully explicit coefficient theorem

Theorem 10.1 (Product-weighted Lagrange–Bürmann). *Let $\Phi(t)$ be a product of the form (4.2), with logarithmic inputs L_j from (4.3). There is a unique $w(z) \in zR[[z]]$ satisfying $w = z\Phi(w)$. For integers $n \geq m \geq 1$,*

$$[z^n]w(z)^m = \frac{m}{n} \mathcal{E}_{n-m}(nL) = \frac{m}{n} \sum_{\mathbf{a} \in \mathcal{P}(n-m)} \prod_{j=1}^{n-m} \frac{(nL_j/j)^{a_j}}{a_j!}. \quad (10.1)$$

For $H(t) = \sum_{r \geq 0} h_r t^r$ and $n \geq 1$,

$$[z^n]H(w(z)) = \frac{1}{n} \sum_{r=1}^n r h_r \mathcal{E}_{n-r}(nL). \quad (10.2)$$

The constant coefficient is h_0 . For $n < m$ the coefficient of w^m is zero.

Proof. The map $t \mapsto t/\Phi(t) = t + O(t^2)$ has a unique formal inverse, which is w . Alternatively, iteration from $w = 0$ improves the z -adic precision at every step and proves existence and uniqueness without analytic assumptions. For the coefficient formula, make the formal residue substitution $z = t/\Phi(t)$:

$$\begin{aligned} [z^n]w^m &= \text{Res}_{t=0} t^{m-n-1} \Phi(t)^n \left(1 - t \frac{\Phi'(t)}{\Phi(t)} \right) dt \\ &= [t^{n-m}] \Phi^n - \frac{1}{n} [t^{n-m-1}] (\Phi^n)' = \frac{m}{n} [t^{n-m}] \Phi^n. \end{aligned}$$

The residue change is valid for a substitution with linear coefficient 1; it can also be checked on each Laurent monomial, whose derivative has residue zero except for the logarithmic differential. Apply [theorem 4.1](#) to the last coefficient. Expanding H and using linearity gives (10.2). $\square$

This is ordinary Lagrange inversion over a coefficient field containing q . Neither the prefactor m/n nor the derivative in Lagrange–Bürmann is replaced by a q -integer or a Jackson derivative.

10.2 A finite Gaussian convolution form

For $p, N \geq 0$ define

$$C_r^{(p,N)}(q) = \sum_{\substack{k_1+\dots+k_p=r \\ 0 \leq k_i \leq N}} q^{\sum_{i=1}^p \binom{k_i}{2}} \prod_{i=1}^p \begin{bmatrix} N \\ k_i \end{bmatrix}_q = [t^r](-t; q)_N^p. \quad (10.3)$$

For $p = 0$ or $N = 0$ this is δ_{r0} . The equality follows by applying the finite theorem to each of the p factors; this definition requires no recurrence.

Corollary 10.2 (A slot-weighted Fuss family). *If $w = z(-w; q)_N$, then*

$$[z^n]w^m = \frac{m}{n} C_{n-m}^{(n,N)}(q) \quad (n \geq m \geq 1). \quad (10.4)$$

At $q = 1$ this is $\frac{m}{n} \binom{Nn}{n-m}$. In particular, for $N \geq 1$,

$$[z^n]w|_{q=1} = \frac{1}{n} \binom{Nn}{n-1} = \frac{1}{(N-1)n+1} \binom{Nn}{n}. \quad (10.5)$$

Proof. Use [theorem 10.1](#) in coefficient-extraction form and then [\(10.3\)](#). At $q = 1$, $(-t; 1)_N = (1+t)^N$; the last equality is an ordinary factorial cancellation. $\square$

Although [\(10.4\)](#) contains m/n , the case $m = 1$ has coefficients in $\mathbb{Z}_{\geq 0}[q]$. A direct interpretation proves this: a rooted tree vertex has N ordered optional slots numbered $0, \dots, N-1$; occupying slot j contributes q^j and attaches a subtree. The root equation is exactly $w = z \prod_{j=0}^{N-1} (1 + q^j w)$. Only finitely many such trees have a fixed number of vertices. Products of these tree series give the corresponding assertion for w^m .

For $N = 2$, ordinary expansion of $(1+t)^n(1+qt)^n$ yields the Narayana formula

$$[z^n]w = \frac{1}{n} \sum_{j=0}^{n-1} \binom{n}{j} \binom{n}{j+1} q^j. \quad (10.6)$$

The first four coefficients, starting at z^1 , are

$$1, \quad 1+q, \quad 1+3q+q^2, \quad 1+6q+6q^2+q^3.$$

These are not obtained by replacing the two ordinary binomials in [\(10.6\)](#) by Gaussian binomials.

For a reciprocal example, take $w = z/(aw; q)_N^r$ with integers $r, N \geq 1$. Then

$$[z^n]w^m = \frac{m}{n} a^{n-m} \sum_{\substack{k_1+\dots+k_{rn}=n-m \\ k_i \geq 0}} \prod_{i=1}^{rn} \begin{bmatrix} N+k_i-1 \\ k_i \end{bmatrix}_q. \quad (10.7)$$

Here [theorem 2.3](#) replaces the finite theorem. Formula [\(10.1\)](#) continues to apply when r is not an integer, whereas [\(10.7\)](#) uses rn separate factors and therefore does not.

11 A genuinely shifted equation: the area-type *q*-Catalan series

The equation in [theorem 10.2](#) contains q -dependent coefficients but no dilation of its unknown series. A true q -functional equation behaves differently. The existence of several inequivalent q -Catalan families is well documented; [\[10\]](#) develops a uniform viewpoint. We give a complete derivation of the particular normalization needed here.

Theorem 11.1 (A quotient solution and a finite coefficient formula). *Over $\mathbb{Q}(q)$ define*

$$U(z) = \sum_{n \geq 0} u_n z^n, \quad u_n = \frac{(-1)^n q^{n(n-1)}}{D_n}. \quad (11.1)$$

Then

$$C(z) = \frac{U(qz)}{U(z)} \quad (11.2)$$

is the unique formal solution with constant coefficient 1 of

$$C(z) = 1 + zC(z)C(qz). \quad (11.3)$$

For each $n \geq 0$, its coefficients have the nonrecursive finite expression

$$[z^n]C(z) = \sum_{k=0}^n q^k u_k \sum_{\mathbf{m} \in \mathcal{P}(n-k)} (-1)^{M(\mathbf{m})} M(\mathbf{m})! \prod_{j=1}^{n-k} \frac{u_j^{m_j}}{m_j!}, \quad M(\mathbf{m}) = \sum_j m_j. \quad (11.4)$$

Proof. The coefficients in (11.1) satisfy, for $n \geq 1$, $(1 - q^n)u_n = -q^{2n-2}u_{n-1}$, as follows immediately by cancelling D_n . Consequently $U(z) - U(qz) = -zU(q^2z)$. Division by the unit $U(z)$ gives

$$\frac{U(qz)}{U(z)} = 1 + z \frac{U(q^2z)}{U(qz)} \frac{U(qz)}{U(z)},$$

which proves (11.3). Its coefficient of degree $n \geq 1$ involves only lower-degree coefficients on the right, so the solution is unique. Finally expand $1/U = \sum_{r \geq 0} (-1)^r (U - 1)^r$ by the ordinary multinomial theorem and multiply by $U(qz)$. This is exactly (11.4). $\square$

The first terms are

$$C(z) = 1 + z + (1 + q)z^2 + (1 + 2q + q^2 + q^3)z^3 + O(z^4). \quad (11.5)$$

Equation (11.3), or its binary-tree interpretation, shows that every coefficient belongs to $\mathbb{Z}_{\geq 0}[q]$. Indeed, attach two ordered subtrees to a root and give the second subtree an extra factor q for each of its vertices. The resulting weight rule is precisely $zC(z)C(qz)$. This proves that singularities in the rational summands of (11.4) cancel, including at $q = 1$. At $q = 1$ the equation is $C = 1 + zC^2$, so the usual Catalan sequence is recovered.

The polynomials in (11.5), the slot family in (10.6), and the Gaussian quotient $\left[\begin{smallmatrix} 2n \\ n \end{smallmatrix} \right]_q / [n+1]_q$ are not interchangeable. For example, their usual Catalan index $n = 2$ gives respectively $1 + q$, $1 + q$, and $1 + q^2$; at $n = 3$ the first two already differ. The implicit equation and normalization, not the name “ q -Catalan”, must select the intended family.

12 A two-variable Gaussian inversion theorem

12.1 A closed coupled example

Theorem 12.1 (Coupled shifted-factorial inverses). *Let $M, N \geq 0$. The system*

$$u = x(-v; q)_M, \quad v = y(-u; q)_N \quad (12.1)$$

has a unique formal solution with $u = x + O((x, y)^2)$ and $v = y + O((x, y)^2)$. For $m, n \geq 1$ and integers $r, s \geq 0$ with $m \geq r, n \geq s$,

$$[x^m y^n] u^r v^s = \frac{ms + nr - rs}{mn} C_{m-r}^{(n, N)}(q) C_{n-s}^{(m, M)}(q), \quad (12.2)$$

where every C is the finite Gaussian sum in (10.3). The coefficient is zero if $m < r$ or $n < s$. On the axes the complete boundary formulas are

$$[x^m y^0] u^r v^s = \delta_{s0} \delta_{mr}, \quad [x^0 y^n] u^r v^s = \delta_{r0} \delta_{ns}. \quad (12.3)$$

Proof. Writing $A(t) = (-t; q)_M$, $B(t) = (-t; q)_N$, eliminate v to get $u = xA(yB(u))$. This equation has a unique solution in $x\mathbb{Q}(q)[[y]][[x]]$; its coefficients define the same bivariate formal series as joint (x, y) -adic iteration. Then $v = yB(u)$. Ordinary Lagrange–Bürmann in x , with outer function $t^r B(t)^s$, gives

$$[x^m y^n] u^r v^s = \frac{1}{m} [t^{m-1} y^{n-s}] (rt^{r-1} B^s + st^r B^{s-1} B') A(yB)^m.$$

Because $[y^{n-s}] A(yB)^m = C_{n-s}^{(m,M)} B^{n-s}$, this becomes

$$\frac{1}{m} C_{n-s}^{(m,M)} [t^{m-r}] (rB^n + stB^{n-1} B').$$

For $n > 0$, differentiation of B^n shows $[t^{m-r}] tB^{n-1} B' = (m-r)C_{m-r}^{(n,N)}/n$. The prefactor therefore equals $(r + s(m-r)/n)/m = (ms + nr - rs)/(mn)$. Terms with a negative extraction degree are absent. Finally $y = 0$ gives $(u, v) = (x, 0)$ and $x = 0$ gives $(u, v) = (0, y)$, proving the boundary formulas. $\square$

For $r = 1, s = 0$, the formula simplifies to

$$[x^m y^n] u = \frac{1}{m} C_{m-1}^{(n,N)}(q) C_n^{(m,M)}(q) \quad (m, n \geq 1). \quad (12.4)$$

At $q = 1$ this is $m^{-1} \binom{Nn}{m-1} \binom{Mm}{n}$. With $M = 1, N = 2$, for instance, $[x^2 y] u = 1 + q$. The positive integer coefficients also have a two-type rooted-tree interpretation: u-vertices have M optional slots for v-subtrees, and v-vertices have N optional slots for u-subtrees, weighted by the slot numbers. Thus the rational prefactor in (12.2) does not signal nonintegral coefficients.

12.2 The Good determinant behind the formula

For a general system $w_i = z_i \Phi_i(\mathbf{w})$ with unit Φ_i , the ordinary multivariate Lagrange–Good formula reads

$$[z^n] H(\mathbf{w}) = [t^n] H(\mathbf{t}) \prod_i \Phi_i(\mathbf{t})^{n_i} \det \left(\delta_{ij} - t_j \partial_{t_j} \log \Phi_i(\mathbf{t}) \right). \quad (12.5)$$

It follows by the formal change of variables $z_i = t_i / \Phi_i(\mathbf{t})$ in the multivariate residue: the displayed matrix is exactly the Jacobian on the logarithmic differentials dt_j / t_j . The residue invariance is the multivariate counterpart of the substitution in theorem 10.1, and the formula is one of the base article’s central tools [1].

For (12.1) the determinant is

$$1 - uv \frac{A'(v)}{A(v)} \frac{B'(u)}{B(u)}.$$

Extracting the two factors gives $1 - (m-r)(n-s)/(mn)$ times the product of the two C coefficients, which is (12.2). The preceding proof independently derives this special case without invoking a multivariate residue theorem.

More generally, a factor $(at^\nu; q^h)_N^\alpha$ has

$$t_j \partial_{t_j} \log(at^\nu; q^h)_N^\alpha = -\alpha \nu_j \sum_{\ell \geq 1} a^\ell S_N(q^{h\ell}) t^{\ell\nu}. \quad (12.6)$$

Here ν is a nonzero nonnegative integer vector. Thus, after imposing a fixed multidegree bound, both the determinant and its product factors have explicit finite expansions: expand the determinant over permutations and expand the exponential of the logarithmic product over multiplicities indexed by (ν, ℓ) . Only those with $\ell\nu \leq \mathbf{n}$ can contribute. This gives a direct route from the source's Good theorem to multivariate q -shifted-factorial identities without changing its Jacobian or introducing unjustified Jackson derivatives.

13 Taylor coefficients at $q = 1$: Bernoulli–Bell formulas

Let the ordinary Bernoulli numbers B_j be fixed by

$$\frac{t}{e^t - 1} = \sum_{j \geq 0} B_j \frac{t^j}{j!}, \quad B_1 = -\frac{1}{2}.$$

These are not the Bell polynomials $B_n(x_1, \dots, x_n)$; arguments distinguish our two conventional notations. The Bernoulli normalization agrees with [7]. Write

$$\Delta_r(n, k) = \sum_{j=1}^n j^r - \sum_{j=1}^k j^r - \sum_{j=1}^{n-k} j^r, \quad K = k(n - k). \quad (13.1)$$

These are finite sums. In particular $\Delta_1 = K$ and $\Delta_2 = K(n + 1)$.

Theorem 13.1 (All exponential-coordinate derivatives at one). *For $0 \leq k \leq n$, define*

$$\kappa_1 = \frac{K}{2}, \quad \kappa_{2r} = \frac{B_{2r}}{2r} \Delta_{2r}(n, k) \quad (r \geq 1), \quad \kappa_{2r+1} = 0 \quad (r \geq 1). \quad (13.2)$$

Then, as a formal Taylor series at $t = 0$,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{e^t} = \binom{n}{k} \exp \left(\sum_{j \geq 1} \kappa_j \frac{t^j}{j!} \right), \quad (13.3)$$

$$\left. \frac{d^r}{dt^r} \left[\begin{matrix} n \\ k \end{matrix} \right]_{e^t} \right|_{t=0} = \binom{n}{k} B_r(\kappa_1, \dots, \kappa_r). \quad (13.4)$$

Equivalently, its coefficient of t^r is

$$\binom{n}{k} \sum_{\substack{a + \sum_{j \geq 1} 2j b_j = r \\ a, b_j \geq 0}} \frac{(K/2)^a}{a!} \prod_{j=1}^{\lfloor r/2 \rfloor} \frac{1}{b_j!} \left(\frac{B_{2j} \Delta_{2j}(n, k)}{2j(2j)!} \right)^{b_j}. \quad (13.5)$$

Proof. Set $A(t) = (e^t - 1)/t$, with $A(0) = 1$. Differentiating its formal logarithm and using the Bernoulli generating function gives

$$\frac{d}{dt} \log A(t) = \frac{e^t}{e^t - 1} - \frac{1}{t} = \frac{1}{2} + \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} t^{2r-1}.$$

Integration with zero constant gives

$$\log A(t) = \frac{t}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} t^{2r}.$$

In the factorial quotient for $\left[\begin{matrix} n \\ k \end{matrix} \right]_{e^t}$, the factors $-jt$ cancel to $\binom{n}{k}$. The remaining logarithm is the sum of $\log A(jt)$ for $1 \leq j \leq n$ minus the two analogous denominator sums. This yields (13.2) and (13.3). Bell expansion proves (13.4), and expanding only its linear and even-degree terms proves (13.5). $\square$

Corollary 13.2 (Ordinary derivatives at one). *Let $s(r, j)$ be the ordinary signed first-kind Stirling numbers defined by $x(x-1)\cdots(x-r+1) = \sum_{j=0}^r s(r, j)x^j$. Then*

$$\left. \frac{d^r}{dq^r} \begin{bmatrix} n \\ k \end{bmatrix} \right|_{q=1} = \binom{n}{k} \sum_{j=0}^r s(r, j) B_j(\kappa_1, \dots, \kappa_j). \quad (13.6)$$

The $j = 0$ term uses $B_0 = 1$.

Proof. For $\Theta = q d/dq$, one has $q^r (d/dq)^r = \Theta(\Theta-1)\cdots(\Theta-r+1) = \sum_j s(r, j)\Theta^j$, as is checked on q^m and then by linearity. At $q = e^t$, $\Theta = d/dt$. Evaluate at $t = 0$ and use (13.4). $\square$

For example,

$$\left. \frac{d}{dq} \begin{bmatrix} n \\ k \end{bmatrix} \right|_{q=1} = \binom{n}{k} \frac{K}{2}, \quad (13.7)$$

$$\left. \frac{d^2}{dq^2} \begin{bmatrix} n \\ k \end{bmatrix} \right|_{q=1} = \binom{n}{k} \left(\frac{K^2}{4} - \frac{K}{2} + \frac{K(n+1)}{12} \right). \quad (13.8)$$

For $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q$, these are 12 and 22, in agreement with (5.4).

There is also a probabilistic interpretation that requires no limit theorem. Choose a k -subset $\{i_1 < \cdots < i_k\}$ of $\{0, \dots, n-1\}$ uniformly and let $X = \sum i_j - \binom{k}{2}$. Its moment generating function is $\begin{bmatrix} n \\ k \end{bmatrix}_{e^t} / \binom{n}{k}$, by (2.2). Thus the κ_j are its cumulants. In particular

$$\mathbb{E}X = \frac{k(n-k)}{2}, \quad \text{Var } X = \frac{k(n-k)(n+1)}{12},$$

and every odd cumulant above the first vanishes. This is the cumulant form of Gaussian reciprocity, since X and $k(n-k) - X$ have the same distribution.

14 Roots of unity and all local derivatives

14.1 Cyclotomic orders and the q -Lucas value

Let $\Phi_d(q)$ denote the d th cyclotomic polynomial.

Proposition 14.1 (Exact cyclotomic factorization). *For $0 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{d=1}^n \Phi_d(q)^{e_d(n,k)}, \quad e_d(n,k) = \left\lfloor \frac{n}{d} \right\rfloor - \left\lfloor \frac{k}{d} \right\rfloor - \left\lfloor \frac{n-k}{d} \right\rfloor \in \{0, 1\}. \quad (14.1)$$

Thus a Gaussian polynomial has at most a simple zero at any primitive root of any fixed order. In particular $e_1 = 0$.

Proof. Factor $1 - q^j = -(q^j - 1) = -\prod_{d|j} \Phi_d(q)$ in the quotient $D_n/(D_k D_{n-k})$. The signs cancel. Counting multiples of d gives the exponent, and $n = k + (n-k)$ shows it is zero or one. $\square$

Theorem 14.2 (q -Lucas evaluation). *Let ζ be a primitive d th root of unity, $d \geq 1$, and write $n = ad + b$, $k = cd + r$ with $0 \leq b, r < d$ and $0 \leq k \leq n$. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_\zeta = \binom{a}{c} \begin{bmatrix} b \\ r \end{bmatrix}_\zeta. \quad (14.2)$$

In particular the value is zero when $r > b$. Equivalently, $\begin{bmatrix} n \\ k \end{bmatrix}_q \equiv \binom{a}{c} \begin{bmatrix} b \\ r \end{bmatrix}_q \pmod{\Phi_d(q)}$.

Proof. At $q = \zeta$, the finite product decomposes into full blocks:

$$(-z; \zeta)_n = (1 - (-z)^d)^a (-z; \zeta)_b.$$

The coefficient of z^{cd+r} on the right is zero if $r > b$; otherwise it is $(-1)^{c(d+1)} \binom{a}{c} \zeta^{\binom{r}{2}} \left[\begin{smallmatrix} b \\ r \end{smallmatrix} \right]_{\zeta}$. On the left it is $\zeta^{\binom{k}{2}} \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{\zeta}$. The phases agree because

$$\zeta^{\binom{cd+r}{2} - \binom{r}{2}} = (-1)^{c(d+1)}.$$

For odd d , the exponent is a multiple of d and both sides are 1; for even d , use $\zeta^{d/2} = -1$ and the parity of $c(cd-1)$. This proves the value identity. Since it holds at every primitive root, the polynomial difference is divisible by the monic polynomial Φ_d over $\mathbb{Z}$. $\square$

The congruence is a standard q -Lucas theorem; treatments and extensions appear in [9, 8]. The proof above includes the phase factor, which is easy to lose when using the finite theorem at even-order roots. For example,

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{-1} = \begin{cases} 0, & n \text{ even and } k \text{ odd,} \\ \left(\left[\begin{smallmatrix} n/2 \\ k/2 \end{smallmatrix} \right] \right), & \text{otherwise.} \end{cases} \quad (14.3)$$

14.2 Regularized logarithms give every derivative

The next theorem refines the value formula to a complete local Taylor series. Only finite sums and products are needed for any prescribed coefficient. It also covers $d = 1$, recovering theorem 13.1.

Fix n, k and define signed multiplicities

$$\varepsilon_j = \mathbf{1}_{j \leq n} - \mathbf{1}_{j \leq k} - \mathbf{1}_{j \leq n-k} \quad (1 \leq j \leq n). \quad (14.4)$$

For an integer $\ell \geq 1$ put

$$g_1 = \frac{1}{2}, \quad g_{2r} = \frac{B_{2r}}{2r} \ (r \geq 1), \quad g_{2r+1} = 0 \ (r \geq 1). \quad (14.5)$$

For $a \neq 1$ define the finite rational function

$$h_\ell(a) = - \sum_{v=1}^{\ell} \left\{ \begin{smallmatrix} \ell \\ v \end{smallmatrix} \right\} (v-1)! \left(\frac{a}{1-a} \right)^v, \quad (14.6)$$

where the braces are *ordinary* second-kind Stirling numbers.

Theorem 14.3 (All-order root-of-unity expansion). *Let ζ be primitive of order d , and let $e = e_d(n, k) \in \{0, 1\}$. Define the nonzero algebraic constant*

$$C_\zeta = (-1)^e \prod_{\substack{1 \leq j \leq n \\ d \nmid j}} j^{\varepsilon_j} \prod_{\substack{1 \leq j \leq n \\ d \nmid j}} (1 - \zeta^j)^{\varepsilon_j}, \quad (14.7)$$

and, for $\ell \geq 1$,

$$K_\ell = \sum_{\substack{1 \leq j \leq n \\ d \nmid j}} \varepsilon_j j^\ell g_\ell + \sum_{\substack{1 \leq j \leq n \\ d \nmid j}} \varepsilon_j j^\ell h_\ell(\zeta^j). \quad (14.8)$$

Then in the local coordinate $q = \zeta e^t$,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{\zeta e^t} = C_\zeta t^e \exp \left(\sum_{\ell \geq 1} K_\ell \frac{t^\ell}{\ell!} \right). \quad (14.9)$$

In particular, if J_j denotes the j th t -derivative at zero, then

$$J_j = \begin{cases} 0, & j < e, \\ C_\zeta \frac{j!}{(j-e)!} B_{j-e}(K_1, \dots, K_{j-e}), & j \geq e. \end{cases} \quad (14.10)$$

Every ordinary q -derivative is therefore given by the finite sum

$$\left. \frac{d^r}{dq^r} \left[\begin{matrix} n \\ k \end{matrix} \right]_q \right|_{q=\zeta} = \zeta^{-r} \sum_{j=0}^r s(r, j) J_j. \quad (14.11)$$

Proof. Use the product $\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \prod_{j=1}^n (1 - q^j)^{\varepsilon_j}$ before taking any logarithm. When $d \mid j$, write

$$1 - e^{jt} = -jt \frac{e^{jt} - 1}{jt}.$$

The total power of t is $\sum_{d \mid j} \varepsilon_j = e$; the constant factors produce the first product in (14.7), including its sign. When $d \nmid j$, factor out the nonzero number $1 - \zeta^j$. After removing these constants and t^e , the remaining product is a unit with constant term 1, so its formal logarithm is legitimate even when $e = 1$.

The ℓ th derivative at zero of $\log((e^t - 1)/t)$ is g_ℓ , by the proof of [theorem 13.1](#). For the other factors,

$$\left. \frac{d^\ell}{dt^\ell} \log(1 - ae^t) \right|_{t=0} = \left(a \frac{d}{da} \right)^\ell \log(1 - a) = h_\ell(a).$$

To verify the last equality without an infinite polylogarithm series, use $(a \partial_a)^\ell = \sum_{v=1}^\ell \{v\}^\ell a^v \partial_a^v$ and $\partial_a^v \log(1 - a) = -(v-1)!/(1-a)^v$. Scaling t by j multiplies the derivative by j^ℓ . Summing the signed multiplicities proves (14.8) and (14.9). Expanding the exponential by Bell polynomials gives (14.10). Finally use $q^r \partial_q^r = \sum_j s(r, j) (q \partial_q)^j$ at $q = \zeta e^t$ to obtain (14.11). $\square$

If $e = 0$, $C_\zeta = \left[\begin{matrix} n \\ k \end{matrix} \right]_\zeta$ and [theorem 14.2](#) supplies its shorter value. If $e = 1$, the value is zero but $\partial_q \left[\begin{matrix} n \\ k \end{matrix} \right]_q \big|_{q=\zeta} = C_\zeta / \zeta \neq 0$. For instance $\left[\begin{matrix} 4 \\ 1 \end{matrix} \right]_q = 1 + q + q^2 + q^3$ has, at $\zeta = -1$, $e = 1$ and $C_{-1} = -2$, giving derivative 2. For $d = 1$, all factors are in the divisible class, $e = 0$, $C_1 = \binom{n}{k}$, and K_ℓ equals (13.2).

This theorem illustrates the full coefficient-calculus chain: cyclotomic factorization isolates the zero, Bernoulli and second-kind Stirling numbers compute the regular logarithmic derivatives, Bell polynomials exponentiate them, and signed first-kind Stirling numbers change from the logarithmic coordinate back to ordinary derivatives. Direct substitution into $D_n/(D_k D_{n-k})$ would instead produce uninformative $0/0$ expressions.

15 Exact geometric extrapolation and mode annihilation

Theorem 15.1 (Gaussian extrapolation weights). *Let $N \geq 0$, with $q \neq 0$ and $q^j \neq 1$ for $1 \leq j \leq N$. Define*

$$w_j^{(N)}(q) = \frac{(-1)^{N-j} q^{\binom{N-j+1}{2}}}{D_j D_{N-j}} = \frac{(-1)^{N-j} q^{\binom{N-j+1}{2}}}{D_N} \left[\begin{matrix} N \\ j \end{matrix} \right]_q, \quad 0 \leq j \leq N. \quad (15.1)$$

Then

$$\sum_{j=0}^N w_j^{(N)}(q) t^j = \frac{\prod_{r=1}^N (t - q^r)}{D_N}, \quad (15.2)$$

and

$$\sum_{j=0}^N w_j^{(N)}(q) q^{j\ell} = \begin{cases} 1, & \ell = 0, \\ 0, & 1 \leq \ell \leq N. \end{cases} \quad (15.3)$$

Consequently, if $a_j = L + \sum_{\ell=1}^N c_\ell q^{j\ell}$, then $L = \sum_{j=0}^N w_j^{(N)} a_j$ exactly.

Proof. The coefficient of t^j in $\prod_{r=1}^N (t - q^r)$ is $(-1)^{N-j}$ times the elementary symmetric polynomial of degree $N - j$ on $q, q^2, \dots, q^N$. By [theorem 2.1](#), this is $(-1)^{N-j} q^{\binom{N-j+1}{2}} [N]_q$. Division by D_N gives (15.2). Set $t = 1$ to obtain weight sum 1, and set $t = q^\ell$ to obtain the zero moments. The final statement follows by linearity. $\square$

Equivalently, these are the Lagrange interpolation weights for evaluating a polynomial at zero from its values at $1, q, \dots, q^N$. Thus any polynomial p of degree at most N satisfies $p(0) = \sum_j w_j^{(N)} p(hq^j)$ for any scalar h . In particular, for dyadic refinement $q = 1/2$ and $N = 2$,

$$(w_0, w_1, w_2) = \left(\frac{1}{3}, -2, \frac{8}{3}\right), \quad L = \frac{1}{3}a_0 - 2a_1 + \frac{8}{3}a_2 \quad (15.4)$$

whenever $a_j = L + c_1 2^{-j} + c_2 4^{-j}$. This is an exact finite-spectrum statement, not an assertion that every convergent approximation sequence admits exact extraction from three entries.

For an analytic $f(h) = \sum_{\ell \geq 0} c_\ell h^\ell$ and arguments inside its disk of convergence, the exact remainder is

$$\sum_{j=0}^N w_j^{(N)} f(hq^j) - f(0) = \sum_{\ell \geq N+1} c_\ell h^\ell \frac{\prod_{r=1}^N (q^\ell - q^r)}{D_N}. \quad (15.5)$$

This follows by applying (15.2) termwise to the finite weighted sum. It implies an $O(h^{N+1})$ error for fixed N, q . It does not claim uniformity as N grows or q approaches a forbidden root.

16 Formal identities, analytic domains, and specialization

16.1 What convergence is, and is not, required

Finite Gaussian identities, triangular transforms, and coefficient extraction in $R[[z]]$ need no analytic convergence. Formal exponentials, logarithms, and divided Lagrange formulas do require the indicated rational denominators. For products with infinitely many geometric factors, two valid settings have been used: coefficientwise q -adic products, or rational logarithmic coefficients such as $S_\infty(q^j) = 1/(1 - q^j)$. Neither should be mislabeled as ordinary z -adic convergence of the finite products.

For a fixed complex $|q| < 1$, $(az; q)_\infty$ converges locally uniformly in z : on each compact set, $\sum_{j \geq 0} |azq^j|$ converges uniformly. Away from its zeros, its reciprocal is analytic. Fractional powers of a unit product have a unique analytic branch near $z = 0$ with value 1, agreeing with the formal power used here. The logarithm computation is justified in a sufficiently small disk by absolute convergence, and the identities elsewhere follow by analytic continuation where both sides are defined. These conditions recover the standard analytic q -binomial theorem [\[3, 4\]](#).

If Φ in [theorem 10.1](#) is analytic near zero and $\Phi(0) = 1$, the analytic inverse-function theorem makes its formal solution an analytic germ. The displayed coefficient formulas are then its

actual Taylor coefficients. No claim is made that they converge past the first singularity of that germ. The two-variable system has the analogous local analytic interpretation because its implicit Jacobian at the origin is the identity.

16.2 A specialization checklist

Polynomial identities such as (2.1), (3.1), (6.6), and (14.2) survive specialization at any q . By contrast, three types of rational formulas need attention. First, interpolation on geometric or q -integer nodes requires distinct nodes; colliding nodes require a confluent limit rather than raw substitution. Second, q -divided-power coordinates need invertible $[j]_q!$, so a coordinate formula can have genuine poles even when a separately defined polynomial family does not. Third, infinite-factor logarithms involve $1 - q^j$ in denominators, and a root-of-unity specialization is not available without a separate limiting argument or a proved cancellation in the final coefficients.

The safe procedure for a Gaussian polynomial is to use its subset expansion, its cyclotomic factorization, or [theorem 14.3](#). The fact that an intermediate formula has 0/0 terms is not evidence that the Gaussian polynomial is undefined. Conversely, polynomiality of a Gaussian coefficient does not regularize every expression built from reciprocals of q -factorials.

17 Implementation, verification, and integration with the source

17.1 Finite algorithms suggested by the formulas

The partition kernel (1.4) uses $p(n)$ multiplicity profiles, where $p(n)$ is the ordinary partition number. It is often much smaller than expanding all ordered compositions, but it is not claimed to be the fastest way to compute long coefficient ranges. Product multiplication and triangular recurrences may be more efficient for bulk computation. An explicit nonrecursive formula and an asymptotically optimal algorithm are different objectives.

For one coefficient, a dependable implementation follows this sequence: construct the finitely many logarithmic inputs needed at that degree; enumerate integer partitions; evaluate the products in exact rational or symbolic arithmetic; and simplify only the completed expression before specializing a potentially singular q . For [theorem 14.3](#), first separate the divisible and nondivisible factors, then form $K_1, \dots, K_r$. No infinite series needs to be stored to obtain a derivative of order r .

The accompanying `verify_identities.py` implements independent checks of the displayed identities. Gaussian polynomials are constructed from subset weights rather than from the identities being tested. Product expansions are compared with partition kernels; fixed-point series are compared with Lagrange and coupled formulas; and polynomial derivatives are compared with the Bernoulli–Bell and regularized root formulas. The archive includes the actual machine-readable test report. The included run passed 2,693 exact assertions in four groups. The finite ranges, exact coefficient fields, and per-identity counts are recorded there; these are finite regression tests, not an unlimited formal verification.

17.2 Normalization audit

Operation	Correct object	Incorrect shortcut
Finite product	$q^{\binom{k}{2}} [N]_q$	Omitting $q^{\binom{k}{2}}$
Gaussian inverse	$(-1)^r q^{\binom{r}{2}} [n]_q$	Replacing the inverse by T_{-c}
Ordinary composition	Ordinary $k!/\prod m_j!$ inside (8.1)	Replacing every profile multiplicity by a Gaussian multinomial
Lagrange–Bürmann	m/n and an ordinary derivative	Replacing n by $[n]_q$ without a new theorem
Newton limit	q -integer nodes tend to integers	Treating geometric nodes as if they did not coalesce
Root-of-unity values	Polynomial evaluation or regularization	Substituting into an uncanceled factorial quotient
q -Catalan family	Its defining equation and normalization	Assuming all Gaussian and area refinements coincide
Infinite product	q -adic, logarithmic, or analytic interpretation	Claiming stabilization in the z -adic topology over $\mathbb{Q}(q)$

17.3 Where the extension attaches

The following crosswalk uses labels observed in the canonical source, not its mutable page or theorem numbers. Labels are pointers to mathematical interfaces, not assertions that the new theorems have existing Lean counterparts.

Base-source interface	Extension developed here
<code>thm:newton-expansion</code>	Two node systems, explicit Gaussian interpolation, and the specified q -Stirling inverse arrays (section 7).
<code>thm:bell-symmetric-functions</code>	Geometric-alphabet specialization and the ramified shifted-factorial Bell kernel (section 4).
<code>thm:merged-binomial-inversion</code>	Gaussian inverse weights and subspace-lattice Möbius interpretation (section 6).
<code>thm:merged-weighted-binomial-translation</code>	The corrected composition law; the ordinary additive law does not persist (section 6).
<code>eq:merged-bell-normalization</code>	Coordinate-specific Bell conversion, with ordinary profile multiplicities retained (section 8).
<code>thm:lagrange-burmann</code>	Recurrence-free product-weighted inverse coefficients and coupled inverses (sections 10 and 12).

17.4 Mathematical and formalization status

The classical finite theorem, Vandermonde and Chu–Vandermonde sums, Gaussian inversion, and q -Lucas evaluation are not presented as discoveries. The master kernel and its applications provide a single reproducible route to many further finite formulas. The root-of-unity derivative theorem is proved here in the precise normalization stated, without claiming historical novelty for its regularization method. All proofs are ordinary mathematical proofs. No Lean files were added or compiled for this extension. Exact finite tests are regression checks, not substitutes for universally quantified proofs.

The formulas also identify a natural formalization order: first polynomial Gaussian identities and denominator-free transforms, then finite partition kernels, then normalized formal-series composition and Lagrange inversion, and finally cyclotomic regularization. This order keeps algebraic prerequisites separate from analytic convergence and from singular coordinate changes.

18 Conclusion

Gaussian binomial coefficients are not merely ornamented binomial coefficients. They are coefficient extractors for geometric alphabets, structure constants for q -factorial convolution, and evaluation matrices for q -integer Newton bases. These roles connect directly with ordinary Bell polynomials, ordinary Stirling numbers, and ordinary Lagrange inversion once the coefficient coordinates are stated correctly.

The central computational principle is

$$\text{geometric product} \longrightarrow \text{finite logarithmic inputs} \longrightarrow \text{Bell partition sum.}$$

For inversion, the same inputs are multiplied by the Lagrange index; for multivariate inversion, they are supplemented by a finite Jacobian determinant. At roots of unity, a preliminary cyclotomic factorization removes the zero before the Bell mechanism is used. This yields explicit formulas for coefficients, compositions, inverse series, and local derivatives while making clear which identities extend polynomially to singular parameter values and which do not.

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